

Assignment AI

Student Records Management System using MongoDB

Introduction

This project demonstrates how MongoDB, a NoSQL database, can be used to efficiently store and manage student records. Unlike traditional relational databases, MongoDB provides flexibility with schema-less design, making it ideal for handling diverse academic data such as student profiles, courses, grades, and attendance.

Objectives

- Implement a database system using MongoDB for student records.
- Showcase CRUD operations (Create, Read, Update, Delete).
- Demonstrate scalability and flexibility of NoSQL.
- Provide a practical example of MongoDB integration with Python.

System Architecture

Architecture Overview

The system consists of the following components:

- Frontend (Optional): Simple web interface or CLI for input.
- Backend Application: Python/Node.js code to interact with MongoDB.
- MongoDB Database: Stores student records in collections.
- API Layer: Provides endpoints for CRUD operations.

Conceptual Flow

User → Application → MongoDB Driver → Database → Response

Technology Stack

- Database: MongoDB
- Programming Language: Python (with PyMongo) or Node.js (with Mongoose)
- Framework (optional): Flask/Django or Express.js
- Deployment: Local MongoDB server or MongoDB Atlas (cloud).

Key Features & Implementation

Key Features

- Schema-less design for flexible data storage.
- CRUD operations for student records.
- Indexing for fast queries.
- Easy integration with web applications.

Implementation Details

- Collections: students, courses, grades.

Sample Document (JSON):

```
{  
  "student_id": "054-82059",  
  "name": "FAIZ FASIAL",  
  "course": "Database Systems",  
  "grades": {"midterm": 85, "final": 90}  
}
```

Python Code Example (PyMongo):

```
from pymongo import MongoClient  
  
client = MongoClient("mongodb://localhost:27017/")  
db = client["university"]  
students = db["students"]  
  
# Insert a record  
students.insert_one({"student_id": "054-82059", "name": "FAIZ  
FAISAL", "course": "Database  
Systems", "grades": {"midterm": 85, "final": 90}})  
  
# Query a record  
for s in students.find({"course": "Database Systems"}):  
    print(s)
```

GitHub Repository

<https://github.com/faizgaba2016-svg/MongoDB-Student-Records-by-FAIZ>